Can Language Models Learn a PDE?

An Interactive Benchmark for Experimental Design
in Weak-Form Coefficient Identification

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Abstract

We introduce an interactive benchmark in which language models choose measurements to identify the coefficients of a one-dimensional elliptic equation. The environment provides access to a numerical solution, budgeted weak-form queries, and free diagnostic tools. Numerical linear algebra is delegated to a solver; the model selects test functions, interprets feedback, and decides when to submit. We reanalyze a public archive of 180 scored runs from six models, three difficulty levels, and ten shared problem seeds per configuration. Recovery rates vary substantially across models and instances, while action counts, within-session improvement, query-family diversity, and confidence reports reveal differences beyond the final error. We regenerate the figures from the archived logs and correct a stale success count in a previously checked-in figure. The diagnostics describe observable behavior rather than validated cognitive faculties. We also distinguish numerical conditioning from structural identifiability: a single solution need not uniquely determine the operator, even within the benchmark's finite-dimensional coefficient class. The archive and reproduction script support verification of the reported aggregates; incomplete attempt histories and provider settings limit broader comparisons.

1 Introduction

An experiment can be difficult because the available measurements are poorly chosen, even when the computations following those measurements are routine. This distinction motivates a benchmark in which a language model directs an inverse-problem investigation. It can request measurements, inspect coefficient estimates, test consistency, and revise its choices. The computational tools perform integration and least-squares recovery, leaving measurement selection and the use of feedback as the model's main responsibilities.

The task builds on the weak formulation of an elliptic boundary value problem [4]. It also connects to active data selection [7]: the informative value of a measurement depends on the measurements already collected. A small residual does not necessarily imply accurate coefficients, and visibly different test functions need not add independent constraints. These features provide concrete opportunities to study how models respond to numerical evidence.

The benchmark was introduced in a Kaggle write-up [8], with behavioral interpretations motivated by the cognitive framework of Burnell et al. [2]. Here we organize the task around three contributions: an instrumented interface for tool-assisted coefficient identification; a reproducible descriptive analysis of 180 archived scored runs; and an account of the numerical and interpretive limits of that analysis. The question in the title concerns identification during a session, not training model weights. The behavioral measurements are hypotheses about useful aspects of scientific problem solving, not independent validation of general cognitive abilities.

Table 1: Actions available in a session. Only stored queries consume the query budget; multiple commands may share a response within a finite turn limit. A free consistency check supplies a measurement but does not append a row to the solve system.

Action	Function
`eval_solution`	Evaluate u and its numerical derivative at chosen points.
`QUERY`	Return r_k and store the corresponding rows of A and F.
`term_integrals`	Inspect the three unweighted terms $\int u'\varphi'$, $\int u'\varphi$, and $\int u\varphi$ before querying.
`solve`	Recover coefficient vectors and report changes from the previous solve.
`check`	Compare a candidate weak-form measurement with the current estimate.
`PREDICT`	Submit the coefficient vectors from the most recent solve.

2 Task and numerical interface

2.1 Hidden equation and measurement system

The hidden boundary value problem is

$$-(a(x)u'(x))' + b(x)u'(x) + c(x)u(x) = f(x), \qquad x \in [0,1], \quad u(0) = u(1) = 0. \tag{1}$$

Each coefficient $q \in \{a, b, c, f\}$ is represented by $q(x) = \sum_{j=0}^{N-1} \theta_{q,j} P_j(x)$, where P_j denotes the ordinary shifted Legendre polynomial on $[0, 1]$. These polynomials are orthogonal, but are not normalized to have unit L^2 norm. The environment samples the coefficients and computes a numerical solution. The model accesses u and u' through an evaluation tool.

For a test function φ_k vanishing at both endpoints, integration by parts gives

$$\int_0^1 au'\varphi_k' \,\mathrm{d}x + \int_0^1 bu'\varphi_k \,\mathrm{d}x + \int_0^1 cu\varphi_k \,\mathrm{d}x = r_k, \qquad r_k = \int_0^1 f\varphi_k \,\mathrm{d}x. \tag{2}$$

Define the basis-weighted matrices

$$\begin{aligned} D_{kj} &= \int_0^1 P_j u'\varphi_k' \,\mathrm{d}x, & B_{kj} &= \int_0^1 P_j u'\varphi_k \,\mathrm{d}x, \\ C_{kj} &= \int_0^1 P_j u\varphi_k \,\mathrm{d}x, & F_{kj} &= \int_0^1 P_j \varphi_k \,\mathrm{d}x. \end{aligned} \tag{3}$$

After K queries, the solver separately minimizes $\|A\theta - r\|_2$, with $A = [D \mid B \mid C] \in \mathbb{R}^{K \times 3N}$, and $\|F\theta_f - r\|_2$. The model chooses measurements; it does not supply its own least-squares implementation. The interface requires at least $4N$ stored queries before solving. This is a protocol rule, not a guarantee of rank or uniqueness.

2.2 Actions and budgets

The protocol permits one stored query per response and prevents submission in the same response as a query. Tool calls and reasoning share a finite turn budget. The common instructions encourage repeated solves and ask for coefficient-wise confidence reports after solving. Consequently, iteration counts reflect behavior under an explicit instruction to iterate.

All archived runs record the `standard` prompt condition. The write-up calls this the enhanced condition: it adds an initial hint about unequal coefficient identifiability. Both prompt variants retain the same equation, tools, and protocol. The recovered archive does not support a baseline-versus-standard ablation. Its logged basis sizes are $2, 3, 4$, correcting the write-up's $3, 5, 8$ description.

Table 2: Settings recorded in the recovered six-model archive. Each row contains ten problem seeds per model.

Difficulty	N	Total coefficients	Queries	Turns
Easy	2	8	12	36
Medium	3	12	18	54
Hard	4	16	24	72

3 Recovery, conditioning, and identifiability

For each coefficient block, the reported error is the absolute maximum coefficient deviation,

$$E_q = \max_{0 \le j < N} |\widehat{\theta}_{q,j} - \theta_{q,j}|, \qquad E = E_a + E_b + E_c + E_f. \tag{4}$$

The main success threshold is $E < 0.01$, with $E < 0.1$ reported as a sensitivity check. These thresholds match the archived scoring and plotting code. They are not thresholds on function error at held-out spatial points, as stated in the original write-up. The scorer separately evaluates function errors on a 500-point grid. Since $|P_j(x)| \le 1$, the total function error is bounded above by NE, but coefficient and function thresholds of the same numerical value are not equivalent.

Recoverability depends on the joint design matrix A, not only on the number or appearance of the test functions. Full column rank gives unique least-squares operator coefficients; small singular values make estimates sensitive to quadrature, discretization, and roundoff. Separate block condition numbers can reveal weak contributions but cannot rule out dependencies *between* blocks. The source is recovered from a different matrix F; its small observed errors should not be attributed to linearity alone, since all four recovery problems are linear in their unknown coefficients.

The reaction weights can be small when u is small, but their relative magnitude is instance-dependent. A large unweighted reaction integral does not ensure that a proposed row adds independent information. Screening term weights is therefore a possible heuristic, not an optimal-design guarantee. Similarly, a low consistency residual can coexist with substantial coefficient error.

Remark 1 (A single solution need not identify the operator)**.** *Take $u(x) = x(1-x)$ and $f(x) = 2$. For any sufficiently small ε, the coefficients*

$$a = 1 + \varepsilon, \qquad b = \varepsilon(4x - 2), \qquad c = -8\varepsilon$$

satisfy the same boundary value problem because

$$2\varepsilon + \varepsilon(4x - 2)(1 - 2x) - 8\varepsilon x(1 - x) = 0.$$

These perturbations lie in the $N = 2$ coefficient class. For example, $\varepsilon = -1/8$ preserves positive diffusion and gives nonnegative reaction. No choice of test functions can distinguish these operators from $(a, b, c) = (1, 0, 0)$ using this solution and source alone.

Thus the benchmark does not have unconditional structural identifiability. Uniqueness results using boundary maps or time-dependent observations [1, 3, 10] concern different observation operators and do not establish it here. A stronger benchmark should characterize joint rank on its sampled instances, include nonidentifiable cases explicitly, and test additional sources or boundary data where uniqueness is required.

Table 3: Exact model IDs in the recovered archive. The names and settings describe the April 2026 experiments, rather than current provider offerings.

Display name	Recorded model ID
Gemini 3.1 Pro	`gemini-3.1-pro-preview`
Claude Opus 4.6	`claude-opus-4-6`
GPT-5.4	`gpt-5.4`
Gemini 3 Flash	`gemini-3-flash-preview`
Claude Sonnet 4	`claude-sonnet-4-20250514`
GPT-5.4 mini	`gpt-5.4-mini`

4 Archived study and provenance

The public Kaggle dataset, *Elliptic Differential Equation LLM Reasoning*, version 1, contains 180 scored run logs and an aggregate summary, released on 16 April 2026. There are six model IDs, three difficulties, and exactly seeds 1–10 in every cell (Table 3). Ground-truth coefficients agree across models for each seed–difficulty pair, permitting comparisons on the same ten tasks. The selected archive contains scored records; it does not establish how many failed or retried attempts preceded them.

OpenAI records specify `reasoning_effort=medium`; other providers have no recorded reasoning-effort override. Temperature is absent from the archived run configuration. A separately published *PDE Library*, version 9, provides an implementation snapshot, but is not pinned per run and postdates some records. We can reproduce aggregates from the saved outputs without claiming to recreate the exact provider calls or every historical source revision.

The original draft also contained a distinct three-model, two-seed pilot. Those tables are preserved in Appendix C; they are not pooled with the recovered study. The proposed five-seed, two-prompt experiment in that draft is not the design of the recovered 180-run archive.

5 Results

5.1 Coefficient recovery

Figure 1 summarizes the scored archive. At $E < 0.01$, Gemini 3.1 Pro succeeds on 10/10, 9/10, and 10/10 easy, medium, and hard instances. Claude Opus 4.6 achieves 10/10, 9/10, and 8/10. GPT-5.4 has a sharper decline, from 7/10 on easy to 3/10 on medium and 1/10 on hard. The remaining models also show substantial task dependence. These are descriptive rates on ten shared instances, conditional on inclusion in the scored archive, rather than estimates of success on all attempted runs.

The regenerated strict-threshold figure corrects GPT-5.4 mini on hard from 0/10 in the checked-in PNG to 2/10. Both the individual logs and archived summary identify seeds 4 and 5 as below 0.01. The corrected count also shows why model-wide rankings and claims about universal difficulty gradients should be avoided. All other displayed success counts agree with the original assets.

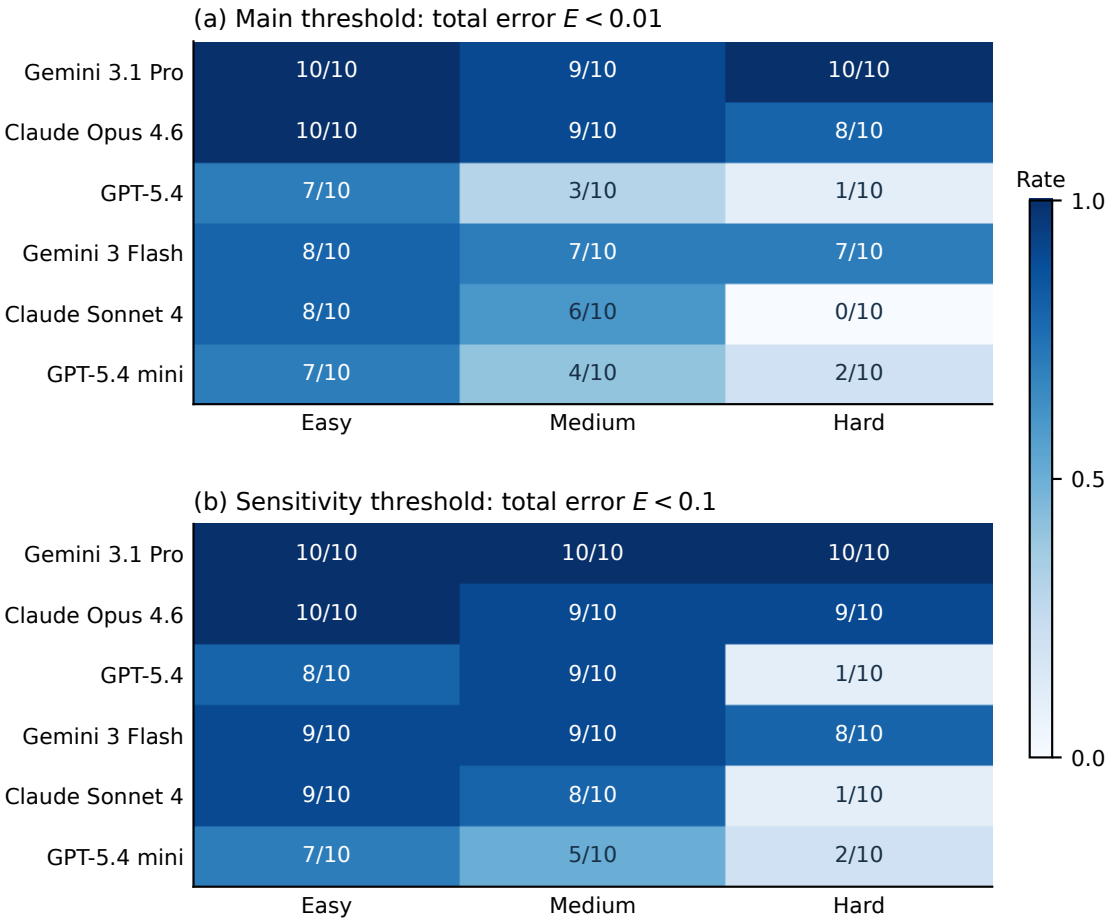


Figure 1: Recovery success regenerated from all 180 archived scored logs. Panels use total absolute coefficient-error thresholds $E < 0.01$ and $E < 0.1$. Each annotation gives successes out of ten scored records. The archive contains no unscored attempts, so these denominators do not measure end-to-end reliability across all attempts. The strict-threshold GPT-5.4 mini hard count is corrected to 2/10.

5.2 Tool use, improvement, and query diversity

The write-up's behavioral figures address useful questions beyond final error, provided their labels are interpreted operationally.

Action use. Figure 2 separates solve calls, consistency checks, and term-integral requests. The profiles show how models allocate tool interactions. More activity does not by itself imply better recovery: the common prompt encourages repeated solves, checks return information without adding solver rows, and efficient strategies can use few actions. The bars stack separate marginal medians, so the height is not necessarily the median total action count.

Improvement during a session. For runs with successive solves, define $R = E_{\text{first}}/E_{\text{last}}$. Figure 3 shows substantial variation across seeds. The original plotting convention assigns $R = 1$ when the improvement ratio is missing, including single-solve runs. A large ratio may reflect a poor initial estimate; it is not evidence of a small final error. Improvement with added measurements also does not alone distinguish model adaptation from routine least-squares improvement.

Query-family diversity. The implementation assigns expressions to five heuristic string-based families: polynomial, trigonometric, exponential, localized, and other. We report

$$H_{\text{norm}} = -\frac{\sum_{i=1}^{5} p_i \log_2 p_i}{\log_2 5},$$

where p_i is the fraction assigned to family i in the archived taxonomy at the final solve, and $0 \log 0 = 0$. Six runs queried again after their final solve; those later queries do not enter this archived entropy. Figure 4 describes differences in this taxonomy. The statistic does not measure linear independence: a diverse family mixture can be redundant, while polynomial-only functions can span useful directions. It should be interpreted alongside the joint matrix and final recovery error.

5.3 Confidence reporting

The interface requests confidence that each coefficient vector is within an absolute error of 0.1. Appendix A examines empirical agreement between these reports and the corresponding recorded estimates. This differs from the write-up's fixed confidence-order diagnostic $\text{Conf}(a) > \text{Conf}(b) > \text{Conf}(c)$, which assumes a hierarchy that need not hold on every instance. It also differs from the legacy design-matrix sensitivity diagnostics discussed in Appendix B.

The calibration analysis uses transcript order to pair reports only with earlier solves, preventing a later improved estimate from being assigned to an earlier confidence statement. Reports with no preceding solve are excluded, and bin counts are retained. Repeated reports from the same run are dependent; the curves are descriptive and are not used to rank general metacognitive ability.

The archive contains 733 reports, of which 707 have a preceding solve and 26 do not. The paired reports contribute 2,121 coefficient observations after excluding the source block. The summary retains counts in disjoint width-0.1 confidence bins. Reports in one response are paired only to strictly earlier solve outputs, rather than a solve requested in that same response. This corrects the older plotting code's forward-looking query-count matching and overlapping bin endpoints.

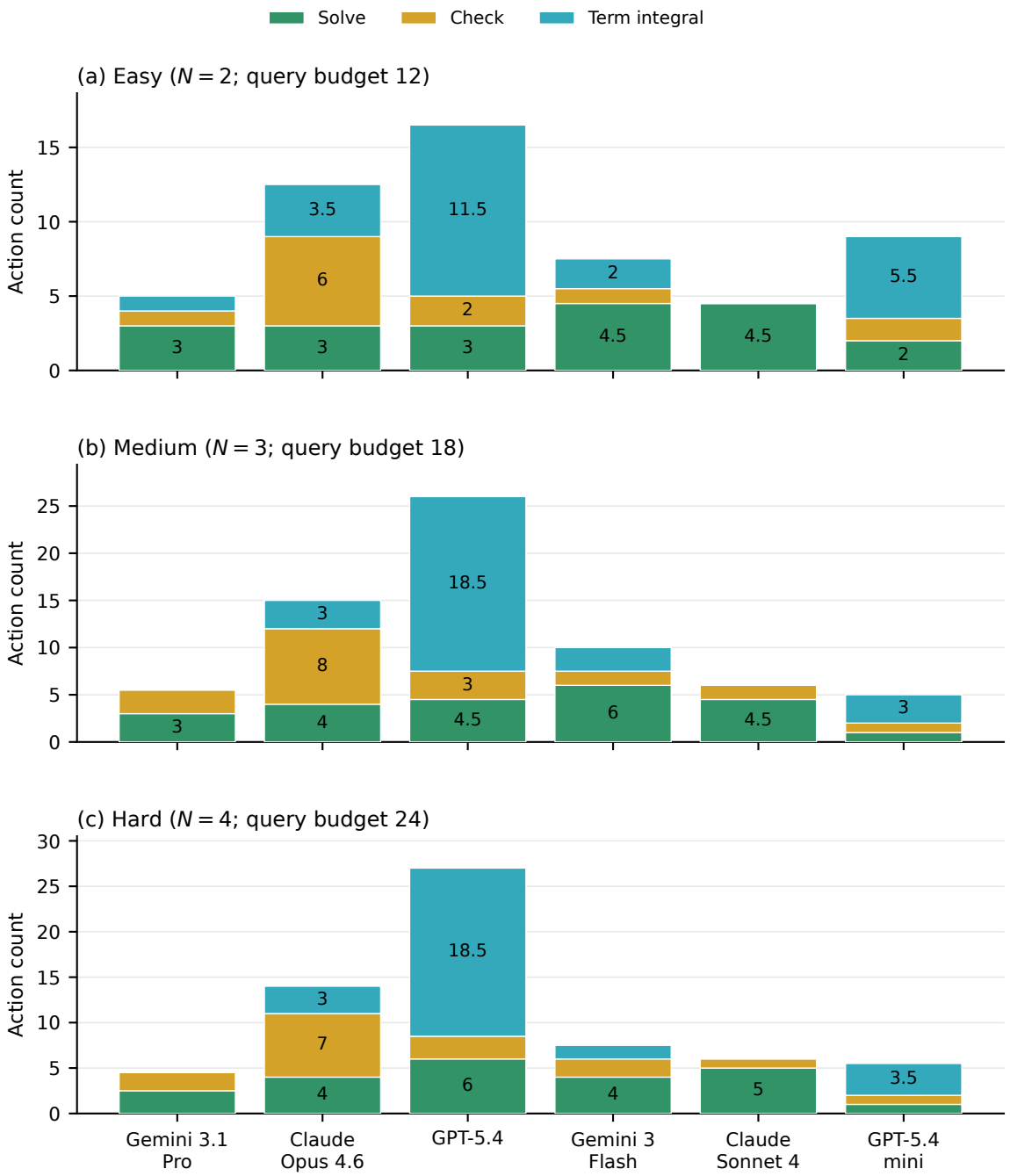


Figure 2: Tool-use profiles regenerated from the archive, with one panel per difficulty. Stacked segments are the separate median counts of solves, consistency checks, and term-integral requests across ten runs per model. The sum of these medians need not equal the median total count. These measurements describe interaction strategies under the supplied protocol.

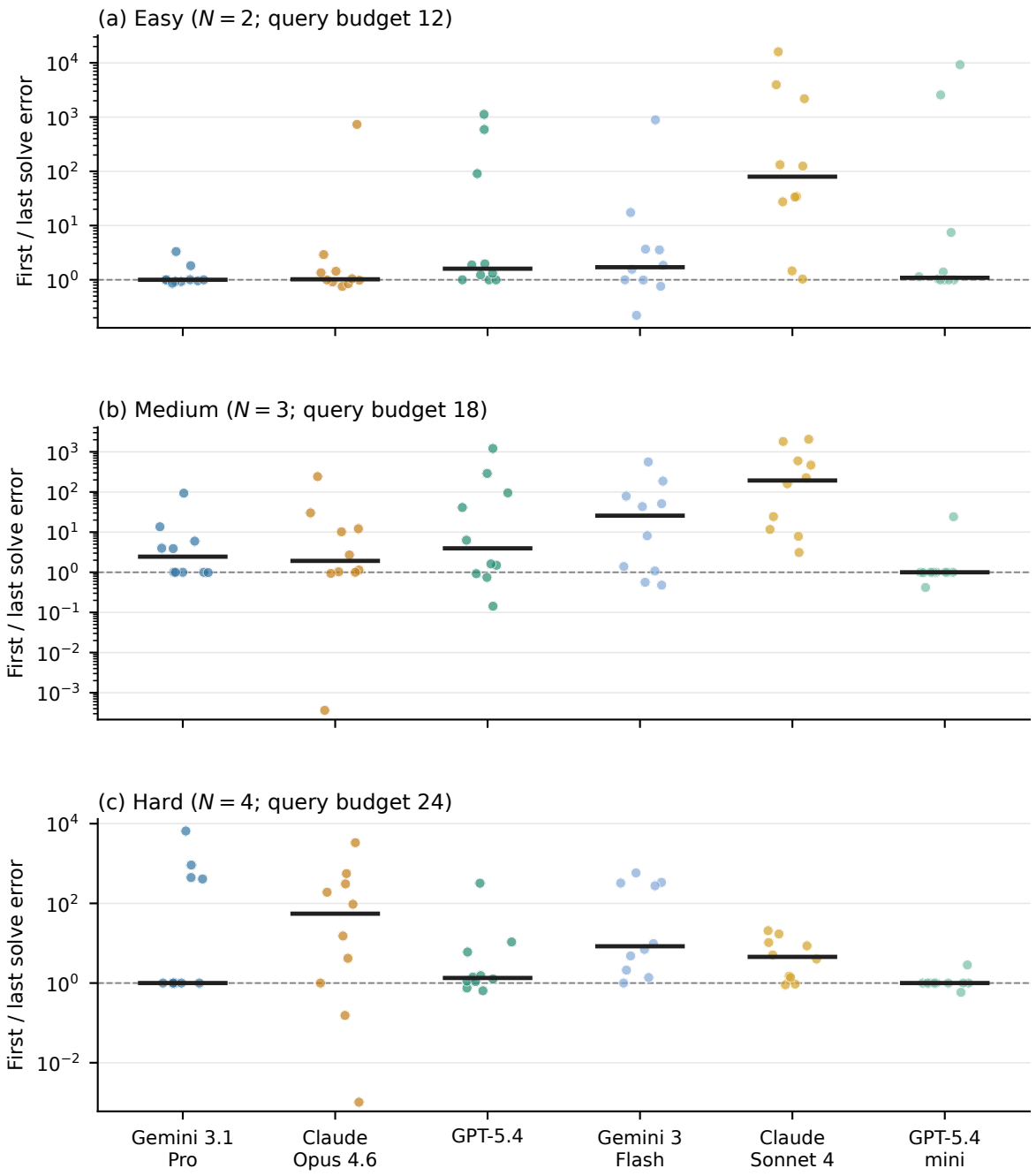


Figure 3: First-to-last solve improvement R, on a logarithmic scale. Dots show individual archived runs and black markers show medians across the ten runs per model and difficulty. Missing ratios receive the historical plotting default $R = 1$. The result measures within-session error change, not model-weight learning or an independently validated learning faculty.

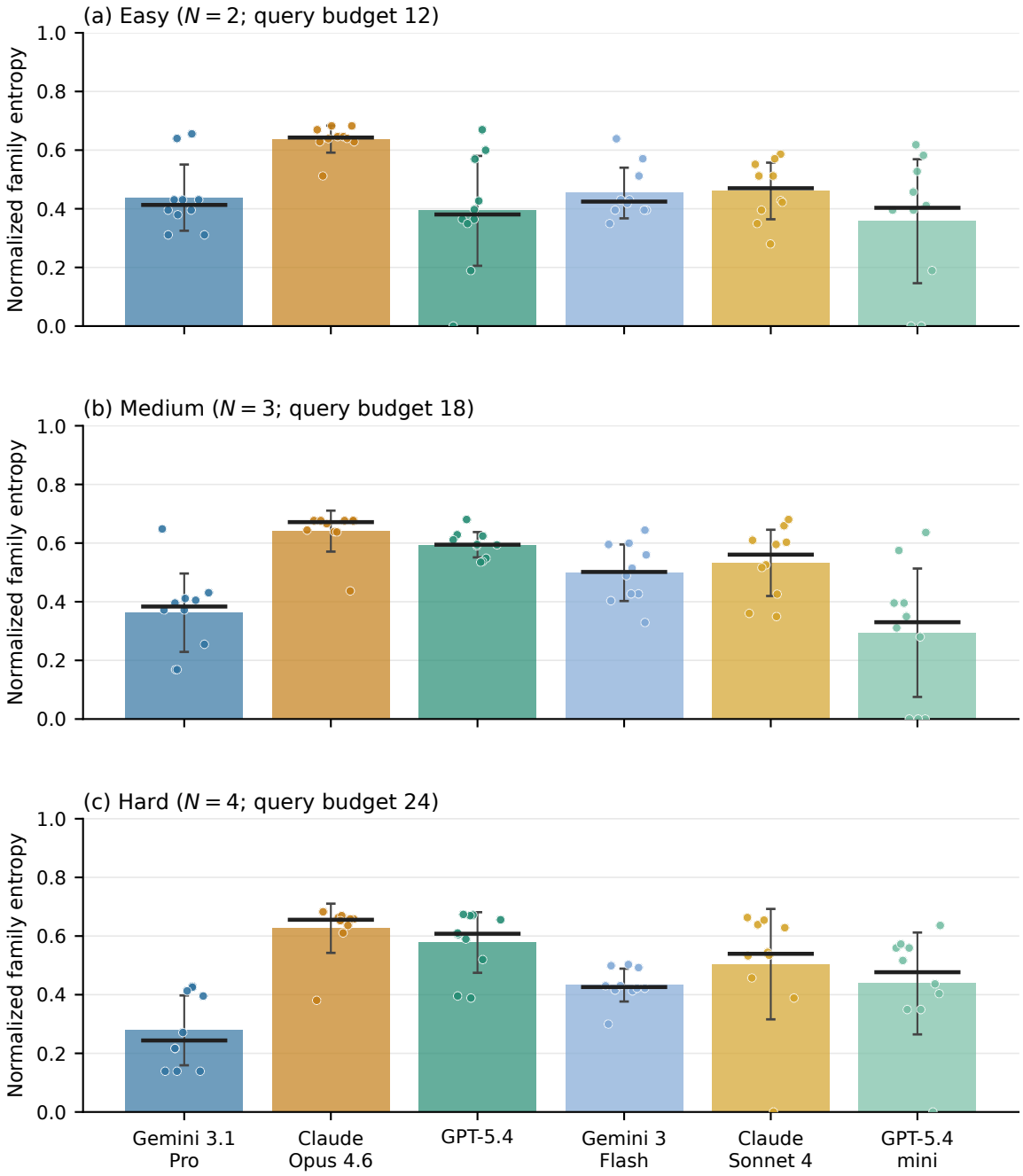


Figure 4: Normalized query-family entropy regenerated from the archive. Bars show means, whiskers one population standard deviation, dots individual runs, and black markers medians. The taxonomy is a heuristic classification of expressions; entropy does not establish informative or independent measurements.

6 Interpretation, limitations, and next experiments

The benchmark isolates an interesting interaction between measurement selection and numerical tools. It is related to applied-mathematics evaluations such as HARDMath2 [9] and mathematical-modeling tasks such as Mamo [6], while emphasizing sequential inverse-problem measurements. Its behavioral axes are inspired by a cognitive taxonomy [2]; they do not establish that the task independently measures eight cognitive faculties.

Several limits matter for interpretation. First, only ten shared PDE seeds per difficulty are represented, and the archive includes scored records rather than a complete attempt history. Second, a text-based interface mixes mathematical decisions with command formatting and protocol compliance. A turn without a recognized command is not equivalent to a rejected mathematical expression: a recognized `QUERY` can still fail validation. Third, repeated solves, confidence reporting, and iteration are explicitly prompted, so action counts cannot establish unprompted planning.

Fourth, single-solution structural nonuniqueness can coexist with numerical conditioning problems. Future evaluations should report joint rank and singular values, distinguish identifiable from ambiguous cases, and compare recovery in coefficient space with out-of-design predictions. Fifth, the archive does not pin complete provider settings or implementation revisions. Reproducing the saved figures is a different claim from reproducing historical model responses.

Useful next comparisons include fixed polynomial and trigonometric query designs, random admissible queries, and adaptive design heuristics under identical query and turn budgets. Baseline and standard prompts should be compared on matched tasks with all attempts retained. Confidence evaluation should define the correctness event in advance, retain report-level sample counts, and account for dependence within runs [5]. These controls would help determine which behavioral differences predict useful experimental decisions.

7 Conclusion

Weak-form coefficient identification supplies a controlled setting for studying how language models choose measurements and use numerical feedback. The recovered 180-run archive permits reproducible comparisons of final errors and concrete interaction patterns. The results support descriptive differences among the evaluated models on these tasks, while the identifiability counterexample and provenance limits constrain stronger claims. The next step is a controlled comparison of measurement strategies with explicit rank checks and complete attempt histories.

Code and data. The implementation is available at https://github.com/Hydrodynamical/PDE_Learning. The Kaggle write-up links the public run archive and implementation snapshot. The repository records source URLs, versions, and archive checksums in `paper_artifacts/data_manifest.json`; the figure-generation script and derived summary are described in `paper_artifacts/README.md`. No new provider calls are required to reproduce this analysis.

A Confidence calibration

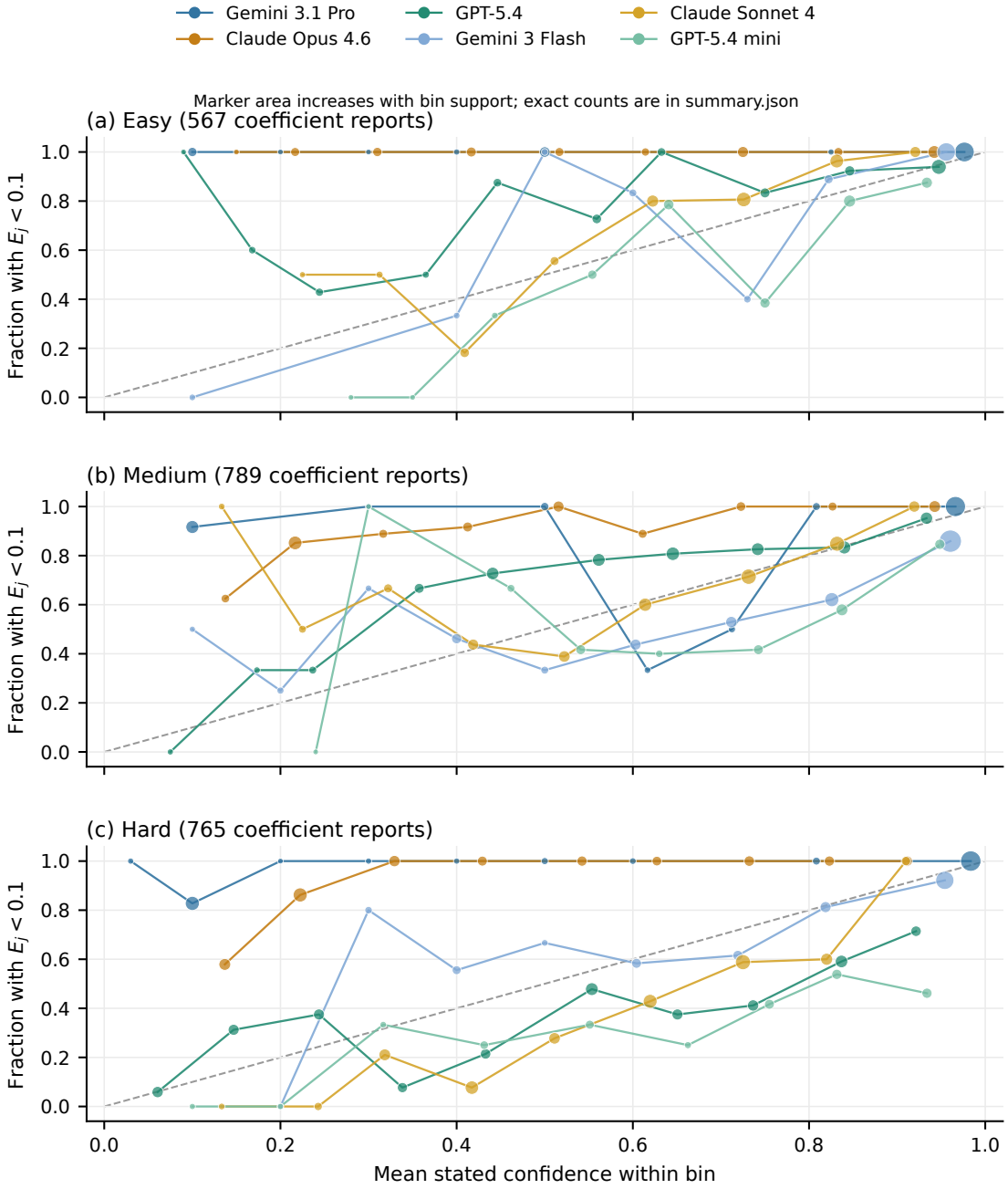


Figure 5: Descriptive confidence calibration with causal report-to-solve pairing. Within each model and difficulty, the fraction of coefficient estimates with $E_q < 0.1$ is plotted against mean stated confidence in disjoint width-0.1 bins. The diagonal denotes agreement. Marker sizes increase with bin support; full counts are saved in the summary. The 707 paired reports contribute 2,121 observations for a, b, c, excluding f. Reports within a run and across coefficient blocks are dependent. No uncertainty bands or general calibration ranking are inferred.

B Legacy sensitivity and monitoring statistics

The archived implementation computes a truncated pseudoinverse H_k of $A_k^\top A_k$, discarding singular values below 10^{-10} times the largest value. It forms block scores

$$s_k(q) = \sqrt{\frac{1}{N}\sum_{i \in q}(H_k)_{ii}}, \qquad q \in \{a, b, c\}.$$

These are design-based sensitivity proxies, not the model's true epistemic uncertainty. Interpreting an inverse information matrix as a covariance requires an appropriate observation-noise model and rank assumptions. The truncated pseudoinverse can assign zero weight to unresolved directions.

The legacy monitoring routine compares rank encodings of s_k and stated uncertainty. Its double-`argsort` ranking does not average tied ranks, so it is not standard tie-aware Spearman correlation. The control routine applies a Sherman–Morrison-style update to the truncated inverse and compares reductions across coefficient blocks for the *chosen* next row. This need not equal a freshly recomputed truncated inverse, and it does not compare alternative query choices. We retain these fields in the source logs but do not use them as calibrated uncertainty measures or as evidence of optimal experimental design.

C Earlier pilot tables

These tables preserve a separate three-model, two-seed pilot whose source logs have not been recovered. The earlier draft described its errors as mean absolute coefficient deviations; that convention cannot be verified and differs from Equation (4). These values should not be pooled or compared numerically with the main study. The legacy $\bar{M}$ column also has the limitations in Appendix B.

Table 4: Legacy pilot table from the earlier draft: three models, three difficulty levels, and two seeds (18 attempts, 15 scored). Values were not regenerated from the six-model archive. "—" indicates the model failed to produce a scored run.

Model	Diff	Seed	E_a	E_b	E_c	E_f	E	Solves	Q used
GPT-4o	easy	42	0.0001	0.0004	0.0026	0.0	0.003	2	11
GPT-4o	easy	137	0.0000	0.0000	0.0000	0.0	0.000	3	12
GPT-5.4-mini	easy	42	0.0000	0.0002	0.0014	0.0	0.002	1	10
GPT-5.4-mini	easy	137	0.0000	0.0000	0.0001	0.0	0.000	1	8
GPT-5.4	easy	42	0.0000	0.0001	0.0004	0.0	0.001	3	12
GPT-5.4	easy	137	0.0000	0.0000	0.0000	0.0	0.000	4	12
GPT-4o	medium	42	—	—	—	—	—	0	9
GPT-4o	medium	137	0.0000	0.0002	0.0019	0.0	0.002	4	18
GPT-5.4-mini	medium	42	0.1861	0.6740	1.4416	0.0	2.302	1	17
GPT-5.4-mini	medium	137	0.0229	0.2777	2.4272	0.0	2.728	3	13
GPT-5.4	medium	42	0.0001	0.0005	0.0009	0.0	0.001	3	17
GPT-5.4	medium	137	0.0000	0.0000	0.0000	0.0	0.000	4	17
GPT-4o	hard	42	—	—	—	—	—	0	0
GPT-4o	hard	137	—	—	—	—	—	0	1
GPT-5.4-mini	hard	42	0.1275	1.4848	0.7268	0.0	2.339	1	20
GPT-5.4-mini	hard	137	0.1196	1.2099	3.5182	0.0	4.848	2	21
GPT-5.4	hard	42	0.0730	2.1305	1.4725	0.0	3.676	7	23
GPT-5.4	hard	137	0.0010	0.0424	0.2458	0.0	0.289	4	24

Table 5: Legacy pilot behavioral summary, preserved from the earlier draft. These medians were not regenerated from the six-model archive. H = normalized family entropy; R = improvement ratio; W = wasted turn fraction; Tail = turns after last solve. Check% = checks as a fraction of queries (checks are free and do not consume budget).

Model	Solves	Budget	Check%	H	R	$\bar{M}$	W	Tail
GPT-4o	3	96%	13%	0.30	1.0	—	0.05	1
GPT-5.4-mini	1	78%	18%	0.00	1.0	0.5	0.08	2
GPT-5.4	4	97%	8%	0.35	4.2	0.8	0.02	1

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